HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process an Airbag Operation

Phase 15: Hardware-Software Integration Testing

Version: 1.00

File	:	Airbag_ECU_Phase_15
Version	:	1.00
Date	:	2026/14/06
Editor	:	Shruti Pawar

Contents

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	11:00	Zoom	ALL
Bericht:	*none*		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklaerung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.

4 Hardware-Software Integration Testing

HW/SW integration testing verifies that real hardware inputs produce correct real outputs by testing the Arduino/test board together with sensors, outputs, and CAN. This section covers the commissioning checklist, pass/fail criteria, fault injection tests, timing tests, and requirement traceability.

4.1 Test Environment and Commissioning Checklist

Tabelle 2 Test Environment and Commissioning Checklist

Item	Expected Setup / Check	Evidence	Status
Controller	Arduino Due or equivalent test board configured with the Airbag ECU prototype software.	Board photo, software version, Git commit ID.	Planned
Analog inputs	ACC on A0, GYRO on A1, PRESSURE on A2; values adjustable from low to high range.	Serial monitor screenshot and measured ADC values.	Planned
Digital input	Baby-seat/passenger suppression switch on D2 with defined ON/OFF state.	Switch state screenshot/log.	Planned
Outputs	Driver airbag indicator D8, passenger airbag indicator D9, warning lamp D13.	LED/output measurement or logic analyzer screenshot.	Planned
CAN	CAN ID 0x120, DLC 8 bytes; normal/crash/fault messages visible.	CAN log or serial-simulated frame printout.	Planned
Power	Stable supply; optional backup capacitor/power-loss simulation if available.	Voltage reading and test notes.	Planned
Safety check	No real pyrotechnic actuator connected. Outputs represented only by LEDs/test loads.	Test setup sign-off.	Planned

4.2 General Pass/Fail Criteria

Tabelle 3 General Pass/Fail Criteria

ID	Pass Condition	Fail Condition
AC-01	No airbag output is activated during normal driving or invalid input conditions.	Any unintended output activation.
AC-02	Driver airbag output activates only for valid crash plus safing condition and no blocking system fault.	Driver output missing during valid crash, or active without valid condition.
AC-03	Passenger airbag output is suppressed whenever baby-seat input is active.	Passenger output ON while baby-seat input is active.
AC-04	Warning lamp activates during detected fault state.	Fault exists but warning lamp remains OFF.
AC-05	CAN frame contains correct status, sensor values, deployment status, fault code, and checksum.	Wrong CAN ID, data length, missing fault code, or checksum mismatch not detected.
AC-06	Output readback matches commanded output, or mismatch causes safe state.	Mismatch exists and system continues normal operation.
AC-07	All safety-relevant requirements have at least one test case and recorded evidence.	Requirement without test evidence.
AC-08	Statement and branch coverage targets are documented and achieved for safety logic.	Coverage missing or untested decision branch remains.

4.3 Fault Injection and Safe-State Tests

Tabelle 4 Fault Injection and Safe State Tests

Test ID	Fault Injected	Method	Expected Safe Reaction	Safety Parameter
FI-01	Sensor too high	Set ACC/GYRO/PRESSURE greater than 950 or invalid range.	Airbags OFF, warning lamp ON, CAN fault sent.	DC / safe state
FI-02	Sensor stuck-at low	Hold sensor at 0 while other inputs change.	No unintended deployment; plausibility fault if implemented.	Fault detection
FI-03	Sensor stuck-at high	Hold sensor above threshold constantly.	No deployment unless full valid crash+safing logic; fault if implausible.	Fault tolerance
FI-04	Output readback mismatch	Force output readback different from command.	Safe state; warning lamp ON.	Output diagnosis
FI-05	Watchdog/program freeze	Stop kicking watchdog or block main loop.	Reset or safe state; fault logged after restart.	Temporal monitoring
FI-06	Memory/config corruption	Modify threshold/config checksum intentionally.	CRC/check failure; safe state.	ROM/config integrity
FI-07	CAN corruption	Wrong ID, wrong DLC, wrong checksum.	Frame rejected and fault reported if required.	Data integrity
FI-08	Power interruption	Simulate power loss or voltage drop.	Backup energy maintains decision/logging if available; otherwise, safe fault.	Availability

4.4 Timing/FTTI Validation Tests

Tabelle 5 Timing/FTTI Validation Tests

Test ID	Scenario	Measurement	Expected Result	Safety Parameter
TIM-01	Loop cycle time	Measure time from sensor read to output decision.	Within project-defined cycle time; no excessive delay.	Deterministic timing
TIM-02	Valid crash reaction time	Measure time from threshold crossing to output activation.	Below chosen FTTI/PST target for prototype.	FTTI/PST
TIM-03	Fault reaction time	Measure time from injected fault to lamp/safe state.	Fault detected before unsafe state can occur.	Fault detection time
TIM-04	CAN update time	Measure time from event to CAN status frame.	Crash/fault frame sent within selected project limit.	Communication timeliness

4.5 Requirement-to-Test Traceability Matrix

Tabelle 6 Requirement to Test Traceability Matrix

Req. ID	Requirement	Test Cases	ASIL	Safety Parameter
SWR-01	Read acceleration sensor input.	SV-01 to SV-05, FI-01 to FI-03	D support	Input validity / diagnostic coverage
SWR-02	Read gyroscope/angular sensor input.	SV-08, SV-09, CR-01 to CR-05	D support	Safing / timing / plausibility
SWR-03	Read pressure sensor input.	SV-06, SV-07, CR-02	D support	Side-impact detection / plausibility
SWR-04	Read passenger/baby-seat digital input.	BS-01 to BS-05	C	Safe input / suppression priority
SWR-05	Activate driver airbag for valid crash.	CR-01, CR-02, CR-05	D	Correct functionality / deterministic logic
SWR-06	Activate passenger airbag only if no baby seat is detected.	BS-01 to BS-04	C	Fail-safe suppression
SWR-07	Activate warning lamp during system fault.	FI-01, FI-04, OUT-03	D support	Fault indication / safe state
SWR-08	Send crash/fault status through CAN.	CAN-01 to CAN-04	D support	Data integrity / communication

Req. ID	Requirement	Test Cases	ASIL	Safety Parameter
SWR-09	Read CAN messages for diagnostic/reset command.	CAN-05, CAN-06	D support	Communication validation / checksum
SWR-10	Read back output status after activation.	OUT-01 to OUT-05, FI-04	D	Output diagnosis
SWR-11	Enter safe state if fault is detected.	FI-01 to FI-08	D	Fail-safe behavior
SWR-12	Support test routines for module/integration testing.	All module, integration, and system tests	D support	V and V / traceability

4.6 Open Issues

Tabelle 7 Open Issues

Issue ID	Current Gap	Impact	Required Action
OI-01	Simple demonstration code does not fully implement physical output readback.	Cannot close SWR-10 with real evidence.	Add readback pins or test-board measurement and safe-state logic.
OI-02	CAN receive and checksum validation are described but not fully implemented in simple code.	CAN-05/CAN-06 cannot pass until implemented.	Add CAN receive handler and checksum verification.
OI-03	Watchdog and memory tests are mentioned but not implemented in simple code.	Timing/memory safety evidence incomplete.	Add watchdog initialization, CRC/config check, and startup self-test.
OI-04	Safing condition is inconsistent in documents: written logic mentions GYRO >= 600 OR ACC >= 700, but code uses only GYRO >= 600.	Test expectation may conflict with implementation.	Decide final safing rule and update both SRS and code consistently.
OI-05	Timing/FTTI target is not numerically defined for the prototype.	TIM tests cannot have clear pass/fail limits.	Define project-level timing target, e.g., max cycle/reaction time for prototype.

5 Figure and table listing

No table of figures entries found.

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Airbag ECU Testing and Validation Plan	V3	31.05.2026	LCP13_Airbag_ECU_Testing_and_Validation_Plan_V3.pdf
2	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruino.docx
3	Airbag ECU Hardware Concept Design Report	V02	22.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

[1] International Organization for Standardization, ISO 26262-3:2018, Road vehicles - Functional safety - Part 3: Concept phase. Geneva: ISO, 2018. https://www.iso.org/standard/68385.html

[2] International Organization for Standardization, ISO 26262-4:2018, Road vehicles - Functional safety - Part 4: Product development at the system level. Geneva: ISO, 2018. https://www.iso.org/standard/68386.html

[3] International Organization for Standardization, ISO 26262-6:2018, Road vehicles - Functional safety - Part 6: Product development at the software level. Geneva: ISO, 2018. https://www.iso.org/standard/68388.html

[4] International Organization for Standardization, ISO 26262-8:2018, Road vehicles - Functional safety - Part 8: Supporting processes. Geneva: ISO, 2018. https://www.iso.org/standard/68390.html

[5] International Electrotechnical Commission, IEC 61508-1:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1. Geneva: IEC, 2010.

[6] International Electrotechnical Commission, IEC 61508-3:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 3: Software requirements. Geneva: IEC, 2010.

[7] ISO/IEC/IEEE, ISO/IEC/IEEE 29119-1:2022, Software and systems engineering - Software testing - Part 1: General concepts. Geneva: ISO, 2022.

[8] International Organization for Standardization, ISO 11898-1:2024, Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical coding sublayer. Geneva: ISO, 2024.

[9] AUTOSAR, Specification of RAM Test, AUTOSAR Classic Platform R24-11. AUTOSAR, 2024. https://www.autosar.org

[10] MISRA Consortium, MISRA C:2025 Guidelines for the use of the C language in critical systems. MISRA, 2025. https://misra.org.uk

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_15	14.06.2026	v1.00 Initial version. HW-SW Integration Testing compiled from Testing Plan V3 Sections 6, 7, 8.6, 8.7, 11, 15.